



# South Africa liquid fuels market story

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Inland supply, corridor capture and tank  
turnover

Durban-Lesedi corridor into Gauteng

8 October 2026

Reported Vopak context and illustrative investment economics; all  
commercial assumptions require validation

Photo source: Royal Vopak | Terminal Durban | vopak.com

Strictly Confidential



# Slow demand growth and greater import reliance set the baseline; road-to-rail shifts, EV adoption, power recovery and refinery changes could reshape the outlook

| Baseline / reference                                                                                                         | What the evidence establishes                                                                             | What changes the outlook                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  <b>Fuel demand and power</b>               | 2013–2023 CAGR: petrol –2.08%; diesel +0.82%. Eskom OCGT fuel use fell 40% in FY2025.                     | Growth, rail, EVs and efficiency alter transport demand; fleet recovery and new generation alter diesel dispatch. |
|  <b>Domestic supply and imports</b>         | 2020–2025 imports rose by 5.43bn litres of diesel and 2.73bn of petrol. The sourcing mix has shifted.     | SAPREF and PetroSA restart paths, Secunda gas/MRG and Natref output can change domestic supply.                   |
|  <b>Corridor position and tank turnover</b> | Durban and Lesedi provide coastal and inland assets serving Gauteng. Working capacity and turns are open. | Route costs, access and customer capture determine flow; achievable turnover determines handling headroom.        |
|  <b>Medium / Medium reference world</b>   | M/M anchors the nine-world matrix. Historical evidence is assembled; numerical calibration remains open.  | Vary demand and supply around M/M, then test capture, turnover and returns to identify investment triggers.       |

Executive summary — baseline and changes. Comparable historical periods differ: Department sales 2013–2023; SARS imports 2020–2025; Eskom FY2025 vs FY2024 (own diesel + kerosene). M/M is a reference framework, not a calibrated forecast.

# Prioritise capture and turnover through existing assets; expand where profitable demand exceeds achievable capacity

| Working recommendation                                                                                                     | Investment logic                                                                                            | Condition for commitment                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  <b>Capture inland supply</b>             | Win accessible customer flows from imported and domestic sources.                                           | Prove delivered competitiveness and sufficient customer volume.                                              |
|  <b>Use existing assets first</b>         | The worked illustration favours existing assets in six worlds; access/dispatch improvement in two.          | Confirm working capacity and the actual operating constraint.                                                |
|  <b>Stage corridor and capacity spend</b> | Only high demand / low domestic supply favours extra tanks, with a narrow advantage over improvement alone. | Fund the least-cost viable option that removes the proven constraint.                                        |
|  <b>Keep strategic options open</b>     | Test all nine worlds and emerging gateways; screen EV adjacencies separately.                               | Validate capture, chargeable revenues and capex first; the illustrated choice reverses under downside tests. |

Working recommendation supported by an authored Lesedi illustration, not a calibrated forecast. Reported capacity is used; capture, working capacity, tariffs, capex and discount rate are assumptions. No customer volume is represented as secured.

# The illustrated expansion case passes narrowly—and depends on capture, revenue and capex

| Decision test                                                                                                 | Numeric condition                                   | ILLUSTRATIVE high / low case                                               | Still needs verification                                                                          |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  <b>Customer volume</b>      | Tanks must outperform improving existing capacity.  | Assumed capture 2.813bn L/year; tanks win above 2.806bn L/year.            | No secured volumes inferred. Validate customer capture and contract duration.                     |
|  <b>Route economics</b>      | Delivered-cost headroom $\geq$ R0/m <sup>3</sup> .  | R750 allowable less R650 assumed logistics = R100/m <sup>3</sup> headroom. | Both prices are authored. Compare actual Durban, Maputo, Walvis Bay and domestic quotes.          |
|  <b>Capacity constraint</b>  | Captured flow exceeds achievable handling.          | Existing 1.890bn L/year; improvement 2.268bn; tanks 2.916bn.               | 90% working capacity and 1.25 / 1.50 monthly turns are assumptions, not reported operations.      |
|  <b>Incremental return</b> | Positive NPV and better value than the alternative. | Improvement NPV R+27m; combined R+35m. Extra advantage only R8m.           | 10% discount rate, 20 years, pre-tax. Actual tariffs, cost, tax and project schedule remain open. |

ILLUSTRATIVE ONLY. 100% project basis, constant ZAR; capex at t0; operating ramp 50%/75%/100%; no tax, working capital, construction delay, inflation or residual value. Turnover and occupancy are distinct. Full assumptions, benchmarks, options and downside tests: pp35–38.

# Only one illustrative world favours tanks; six favour existing assets and two favour improvement

| Gauteng demand / domestic supply | HIGH deliveries<br>4.8bn L/year                                               | MEDIUM deliveries<br>3.6bn L/year                                                        | LOW deliveries<br>2.4bn L/year                                                       |
|----------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| LOW demand<br>6.0bn L/year       | L / H<br>USE EXISTING ASSETS<br>Capture 0.99bn L/year<br>Incremental NPV R+0m | L / M<br>USE EXISTING ASSETS<br>Capture 1.50bn L/year<br>Incremental NPV R+0m            | L / L<br>USE EXISTING ASSETS<br>Capture 2.01bn L/year<br>Incremental NPV R+0m        |
| MEDIUM demand<br>6.8bn L/year    | M / H<br>USE EXISTING ASSETS<br>Capture 1.39bn L/year<br>Incremental NPV R+0m | M / M   BASELINE<br>USE EXISTING ASSETS<br>Capture 1.90bn L/year<br>Incremental NPV R+0m | M / L<br>IMPROVE ACCESS / DISPATCH<br>Capture 2.41bn L/year<br>Incremental NPV R+27m |
| HIGH demand<br>7.6bn L/year      | H / H<br>USE EXISTING ASSETS<br>Capture 1.80bn L/year<br>Incremental NPV R+0m | H / M<br>IMPROVE ACCESS / DISPATCH<br>Capture 2.30bn L/year<br>Incremental NPV R+27m     | H / L<br>TEST IMPROVEMENT + TANKS<br>Capture 2.81bn L/year<br>Incremental NPV R+35m  |

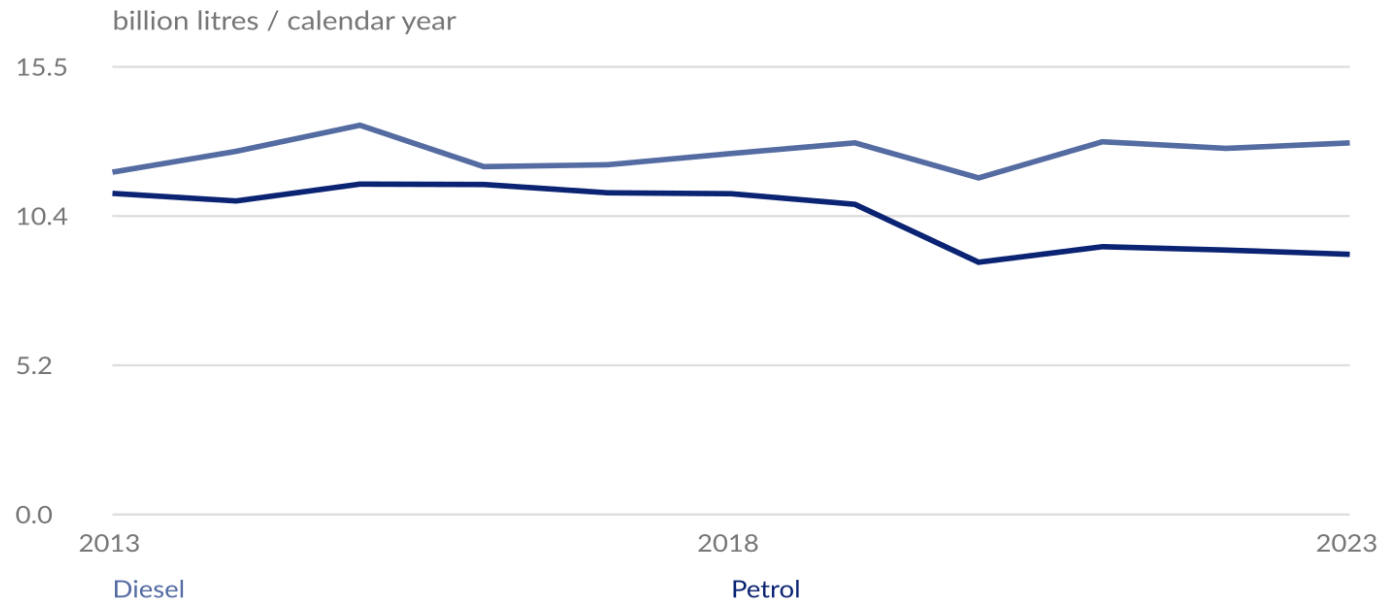
Highest incremental NPV among three illustrative packages; R0 means retain existing assets, not zero terminal value.

ILLUSTRATIVE snapshot, not forecast or probabilities. Domestic supply here means assumed deliveries into Gauteng, not national refinery capacity. Shares, rates and costs held fixed across worlds. Rail/EV opportunities remain separate cases, not valued here. Each cell is relative to existing assets; unserved volumes can remain. Inputs: p37; downside: p38.

# Petrol demand has declined while diesel has grown slowly, limiting the case for demand-led expansion outlook

## Demand trend

## Change over 2013–2023



### 01 | Petrol 2013–2023

11.15 to 9.04 bn litres  
-2.12 bn L / -19.0%  
CAGR -2.1% per year

### 02 | Diesel 2013–2023

11.89 to 12.91 bn litres  
+1.02 bn L / +8.6%  
CAGR +0.8% per year

FIASA 2024 observations are excluded from this comparison: source definitions are not reconciled.

The demand history supports a selective investment case: growth must translate into captured corridor throughput and sufficient tank turns.

Department sales 2013–2023. CAGR from comparable endpoints: petrol -2.08%, diesel +0.82%. FIASA 2024 is excluded from the trend calculation; its observations remain in the evidence data.

# Power shortages and road freight have supported diesel consumption, but that support can change with increased support for rail reform and improved Eskom

## Demand support

### 2024 vs 2025

■ 2024 ■ 2025

### Implication for demand



#### Power generation

Own OCGT fuel • million litres • financial years

OCGT fuel



-39.9%

450m fewer litres in one year. Sustained recovery reduces diesel demand; retirements and replacement timing could reverse it.

EAF: 54.56% → 60.60% | Cost\*: R23.87bn → R13.32bn



#### Road freight

Reported payload • million tonnes • calendar years

Road



-0.3%

Rail



+4.7%

Road still carries 85.3% of tonnes. Rail recovery reduces diesel use only where freight transfers and net fuel savings follow.



#### Sector activity

Real GVA • Rbn, 2015 prices • calendar years

Agriculture



+17.4%

Mining



+0.2%

Manufacturing



-1.2%

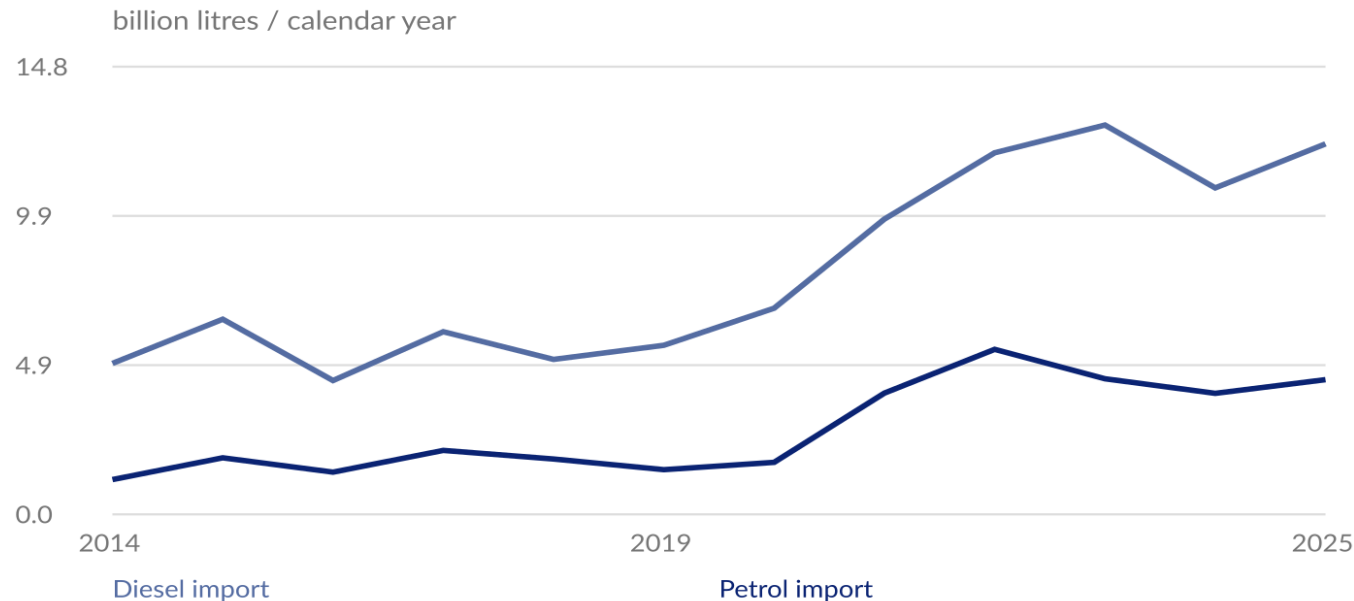
Agriculture grew; mining was flat and manufacturing declined. Sector activity matters more than one GDP assumption.

Eskom Integrated Report 2025 pp100,141: own-fleet diesel + kerosene; \*cost includes storage/demurrage. Stats SA P7162 Dec 2025 p7: both years from the revised vintage. Stats SA P0441 Q2 2026 annual real GVA: agriculture includes forestry/fishing. FY and calendar-year periods shown separately. Bars start at zero; scales are common within each panel, not across different units. Light blue = 2024; navy = 2025.

# The shut down of domestic refining capacity has lowered domestic production and has increased imports shifting how inland demand is supplied

## Import trend

## Change since 2020



### 01 | Diesel import

6.81 to 12.25 bn litres  
Increase: 5.43 bn litres  
CAGR: 12.4% (2020–2025)

### 02 | Petrol import

1.72 to 4.45 bn litres  
Increase: 2.73 bn litres  
CAGR: 21.0% (2020–2025)

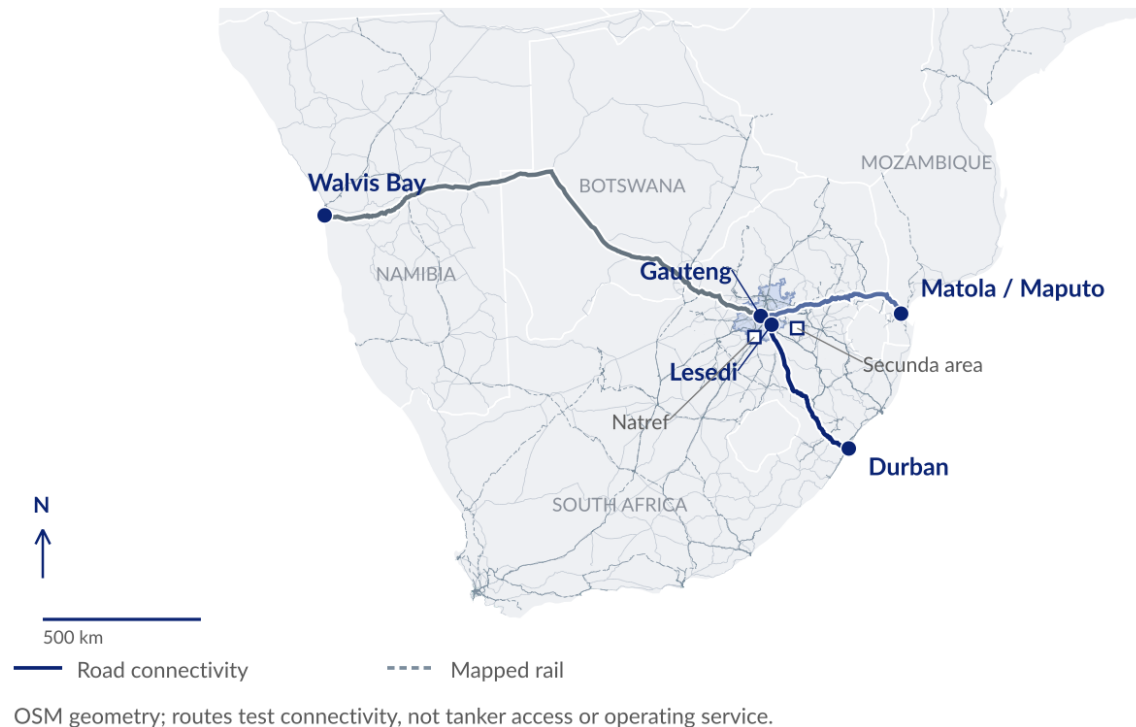
More inland handling opportunity;  
capture and turnover determine benefit.

Additional imported litres create handling demand. Vopak benefits where it captures inland flows and can increase turnover or profitably remove a constraint.

SARS litre-reported petrol/diesel codes, complete calendar years. 2020–2025 changes are historical comparisons, not plant-level causal attribution. Export histories retained in the source report; unit/blend exclusions remain.

# Emerging Maputo and Walvis Bay routes must be tested alongside Durban and domestic supply

## Competing supply corridors



## Same product, same destination

### 01 | Compare delivered cost

Product, handling, transport, border, losses and inventory finance.

### 02 | Establish price headroom

Bearable charge less actual logistics cost. Capacity and reliability constrain capture.

### 03 | Include domestic sources

Secunda and Natref also serve inland demand; routes and access need validation.

### Next action

Match route costs and access to the same product and Gauteng destination.

Maputo and Walvis Bay are emerging alternatives to include by destination. Compare delivered cost, capacity and reliable access before estimating how much inland demand Vopak can capture.

© OpenStreetMap contributors (ODbL; [osm.org/copyright](https://osm.org/copyright)) / Geofabrik; Natural Earth; geoBoundaries. Shared LAEA projection. Paths test road connectivity, not tanker access. Secunda uses a city reference; Natref a mapped site. Terminal access, turn restrictions and operating rail/pipeline services remain unvalidated. Emerging alternatives are in scope; relative competitiveness is not yet quantified.

# Durban and Lesedi connect coastal supply to Gauteng; operating data must establish room for growth

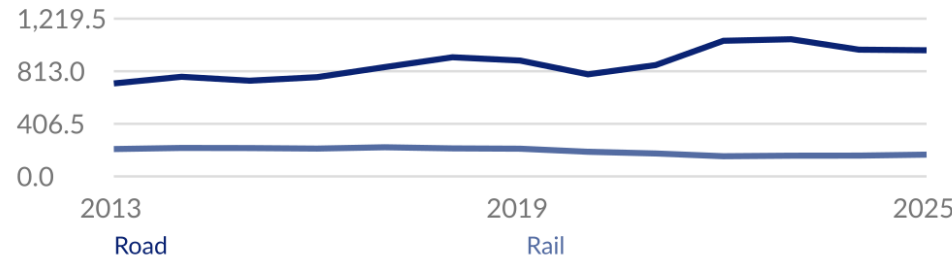
| Asset / measure               | Established position                                                                                                        | What must be established                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Durban</b>                 | 360,246 m <sup>3</sup> published gross capacity.<br>Coastal receipt and onward distribution; mixed products.                | Isolate petrol/diesel working tankage and actual throughput by product.                                          |
| <b>Lesedi</b>                 | 140,000 m <sup>3</sup> published capacity for clean petroleum products.<br>Inland position connected to Durban by pipeline. | Confirm usable working capacity, annual deliveries and monthly tank turns.                                       |
| <b>Operating constraints</b>  | Storage capacity establishes the asset footprint.<br>It does not establish annual handling capacity.                        | Measure receipt, pipeline and dispatch limits, seasonal peaks and stock requirements.                            |
| <b>Implication for growth</b> | Actual throughput and achievable turns are not yet established in this review.                                              | No spare capacity is claimed. Test captured demand against verified handling limits before selecting investment. |

Sources: Vopak terminal directory; Annual Report 2025 confirms Lesedi expansion commissioned in 2025. Published capacity is not working capacity or throughput. Operating data request: p16. Illustrative turnover sensitivity moved to p39; it is not observed utilisation.

# Accelerated rail recovery, EV adoption and ICE technology efficiency could reduce transport fuel demand in the medium to long term

## Rail recovery

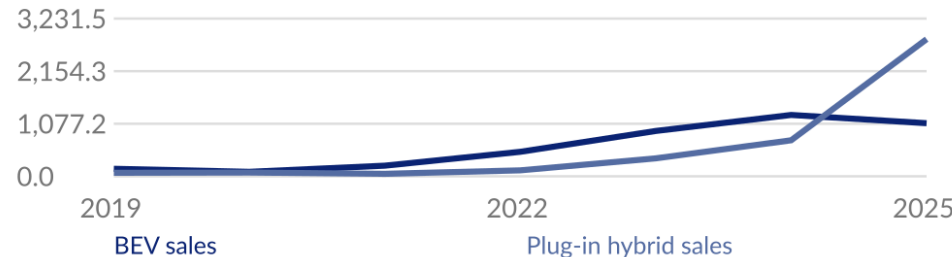
million tonnes



D2: capacity, reliability, transferable freight  
Tonnes require haul distance and traction  
before conversion to net diesel savings.

## EV uptake

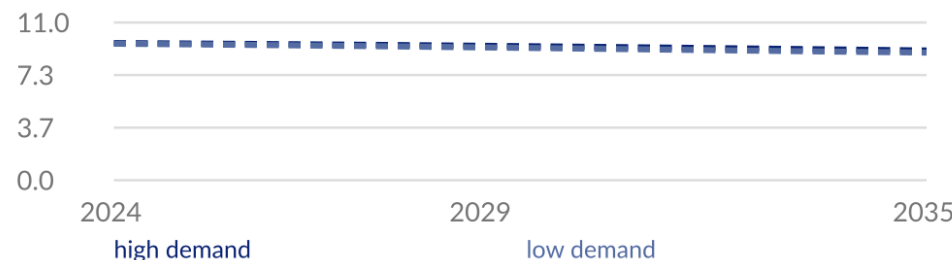
vehicles sold / year



D3: affordability, charging, fleet replacement  
Sales history is not fleet penetration.  
L/M/H adoption calibration remains open.

## Fleet efficiency

L/100 km; draft engine

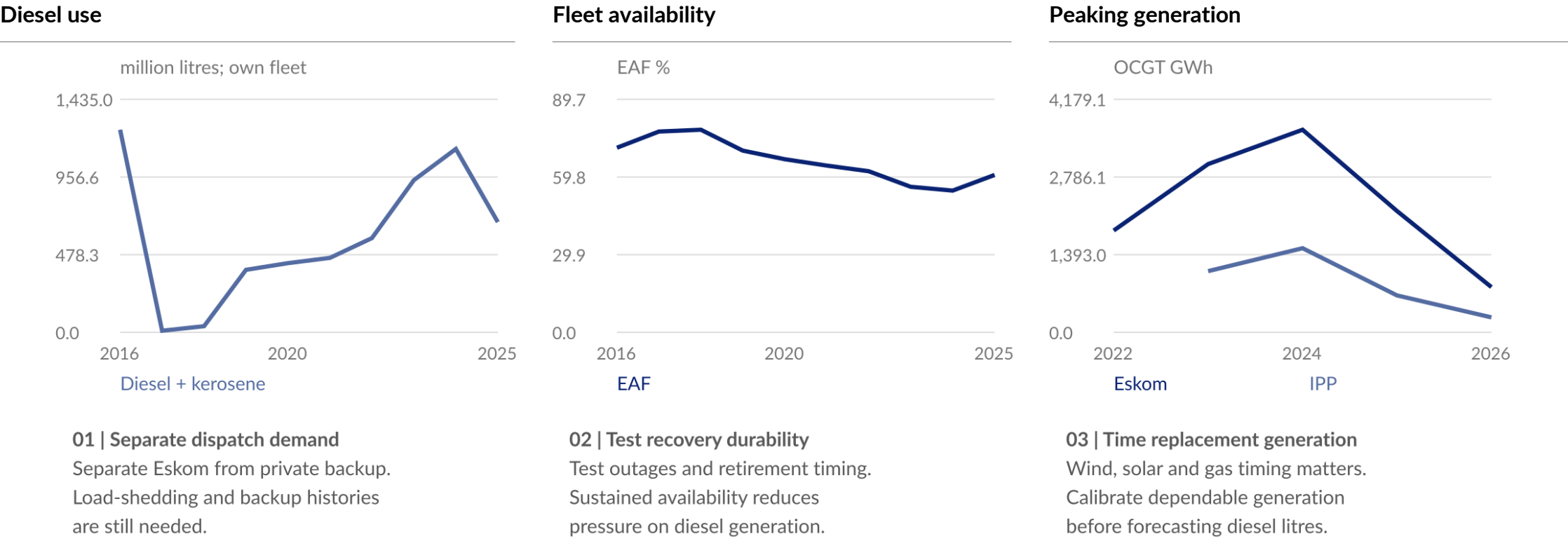


D4: new-cohort efficiency and mileage  
Dashed paths are existing model diagnostics.  
OEM history and Medium calibration are open.

The time series establish the starting points. Rail, adoption and efficiency levers must be calibrated into the same three demand paths before the matrix is quantified.

Stats SA freight payload; staged naamsa BEV/PHEV annual sales 2019–2025; existing engine petrol-ICE efficiency diagnostics. Dashed efficiency paths are assumptions, not historical OEM evidence. Medium and causal fuel impacts remain uncalibrated.

# Power generation recovery could lower diesel use further, while outages and replacement delays could sustain domestic demand



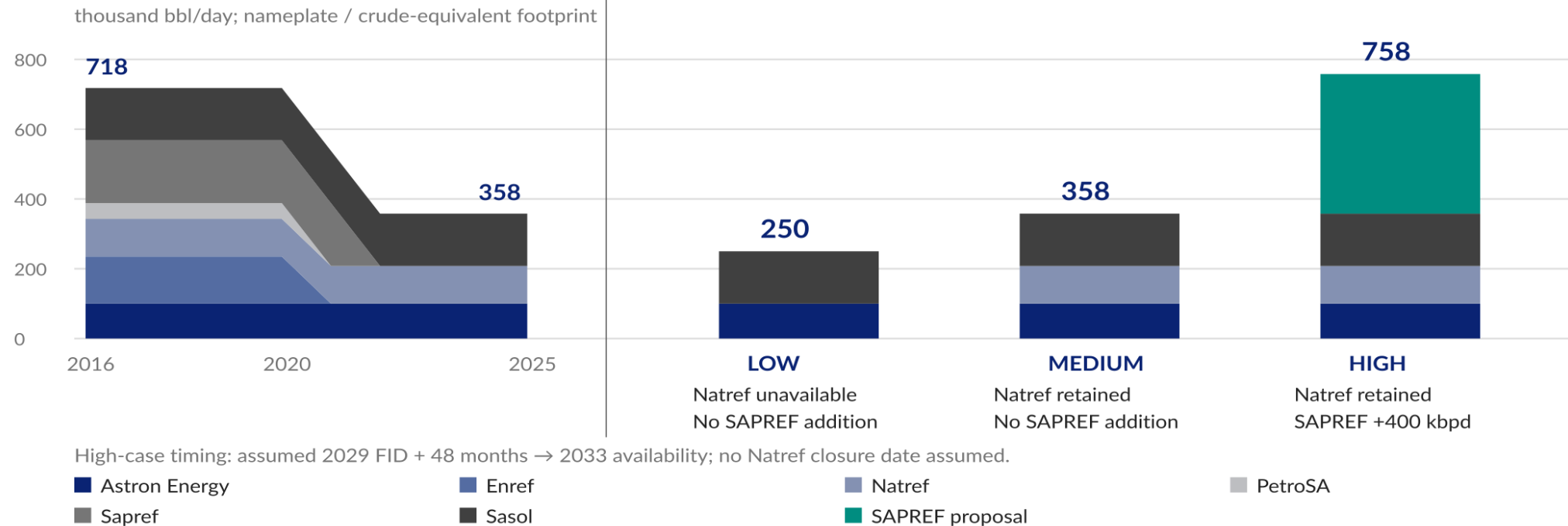
Eskom recovery has lowered diesel dispatch. Wind, solar and gas timing must be tested against demand, fleet retirements and dependable output before forecasting residual diesel.

Eskom FY ending March. Fuel/EAF: 2025 report; GWh: staged 2024–2026 reports. Own Eskom and IPP generation separated. Private backup and load-shedding histories are missing; no assumed annual generation gap is assigned entirely to diesel.

# SAPREF could expand the supply footprint, while Natref unavailability would reduce it

## Reported history

## Illustrative 2036 capacity cases



These capacity cases anchor the supply-axis discussion. Translate them into product output, then test imports, corridor capture and required tank turns in each of the nine worlds.

FIASA 2025 p49; registered review\_capacity\_scenarios.yaml and review\_supply\_worlds.yaml. Illustrative 2036 sensitivities, not calibrated production forecasts. Other plants held at 2025 reported values; Secunda gas/MRG and PetroSA restart effects remain unquantified. Low is not a Natref closure prediction.

## Emerging gateways could redirect inland growth; delivered cost and reliable access determine capture

| Competing route         | What changes the choice                                                | Investment implication                                                             |
|-------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Durban–Lesedi</b>    | Pipeline, rail and road costs; reliable receipt and inland dispatch.   | Test access and handling improvements that protect competitive customer supply.    |
| <b>Maputo / Matola</b>  | Delivered cost, border time and capacity into each inland destination. | Include as an emerging alternative; test customer exposure and corridor responses. |
| <b>Walvis Bay</b>       | Distance, transit, border costs and feasible destination coverage.     | Consider emerging inland opportunities; compare reach customer by customer.        |
| <b>Domestic sources</b> | Secunda / Natref availability, product output and distribution access. | Test inland handling and connections as well as import infrastructure.             |

Compare common product, destination, price date and tax basis. Emerging gateways must be considered; no relative cost ranking is asserted without quotes and verified access. Landlocked SADC opportunities require separate market/access evidence.

# Choose the investment that removes the proven constraint—and earns a return

| Investment option                                                                                                          | When it becomes relevant                                                                          | Decision / commitment trigger                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|  <b>Existing assets / debottlenecking</b> | Customer growth can be served through better tank turns, scheduling or receipt/dispatch.          | Use available working capacity; invest in the specific bottleneck where incremental value is positive.             |
|  <b>Corridor / rail access</b>            | Competitive customer flows are constrained by transport capacity, cost or reliability.            | Secure service and volume commitments. Compare access contracts, loading assets and partnership investment.        |
|  <b>Additional tank capacity</b>          | Profitable captured flows exceed achievable handling and storage remains the binding constraint.  | Size and stage tanks against contracted demand; pass the four tests and downside cases.                            |
|  <b>EV-related adjacencies</b>          | Identified charging / fleet-energy customers offer a viable opportunity as transport electrifies. | Screen partnerships or dedicated assets separately: demand, power access, strategic fit and return must be proven. |

Investment menu, not approved projects. Network rail recovery is an external fuel-demand driver; Vopak rail loading/access is a distinct investment option. EV uptake reduces some fuel demand; it does not itself prove an EV investment case. Project costs, contracts and returns remain open. Three fuel-logistics packages are priced illustratively on p35; these do not constitute engineered rail or EV project estimates.

# Resolve customer volume, working capacity and project economics first to determine the investment

| Data request                                     | Required detail                                                                                                      | Decision it unlocks                                                                                                              |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>FIRST   Volume</b><br>DR02 / DR05             | Monthly unique deliveries, customer destination, contract term and contestable volumes; separate terminal transfers. | Test the illustrated 2.21bn L improvement trigger and 2.806bn L tank trigger against actual customer commitments.                |
| <b>FIRST   Capacity</b><br>DR03                  | Eligible working m <sup>3</sup> , actual turns, committed capacity, stock policy and receipt/dispatch limits.        | Calculate the throughput shortfall and identify whether tanks, access or dispatch bind.                                          |
| <b>FIRST   Economics</b><br>DR04 / DR06          | Matched delivered costs, access, rates, incremental costs and capex; hurdle, ramp-up and working capital.            | Replace the authored rent, chargeable-service margin and capex; these determine whether the narrow expansion advantage survives. |
| <b>THEN   Stress tests</b><br>DR01 / DR07 / DR08 | Reconciled product balance; annual demand, power and plant-output paths.                                             | Test how far demand or supply must change before the preferred investment changes.                                               |
| <b>SEPARATE   EV case</b><br>DR09                | Customer energy demand, charging utilisation, power connection, delivery partner, capex and contribution.            | Decide whether an EV adjacency merits development on its own economics.                                                          |

Request prepared, not sent. Sequence is proposed, not a quantified sensitivity ranking. Nigel to coordinate Vopak data; Manish to reconcile public evidence. Latest 36 complete months where available. Detailed requests and decision triggers logged; owners and commercial inputs remain to be confirmed.

# Appendix

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## Supporting evidence and calibration

Durban-Lesedi corridor into Gauteng

8 October 2026

Detailed evidence, assumptions and remaining calibration requirements

Photo source: Royal Vopak | Terminal Durban | vopak.com



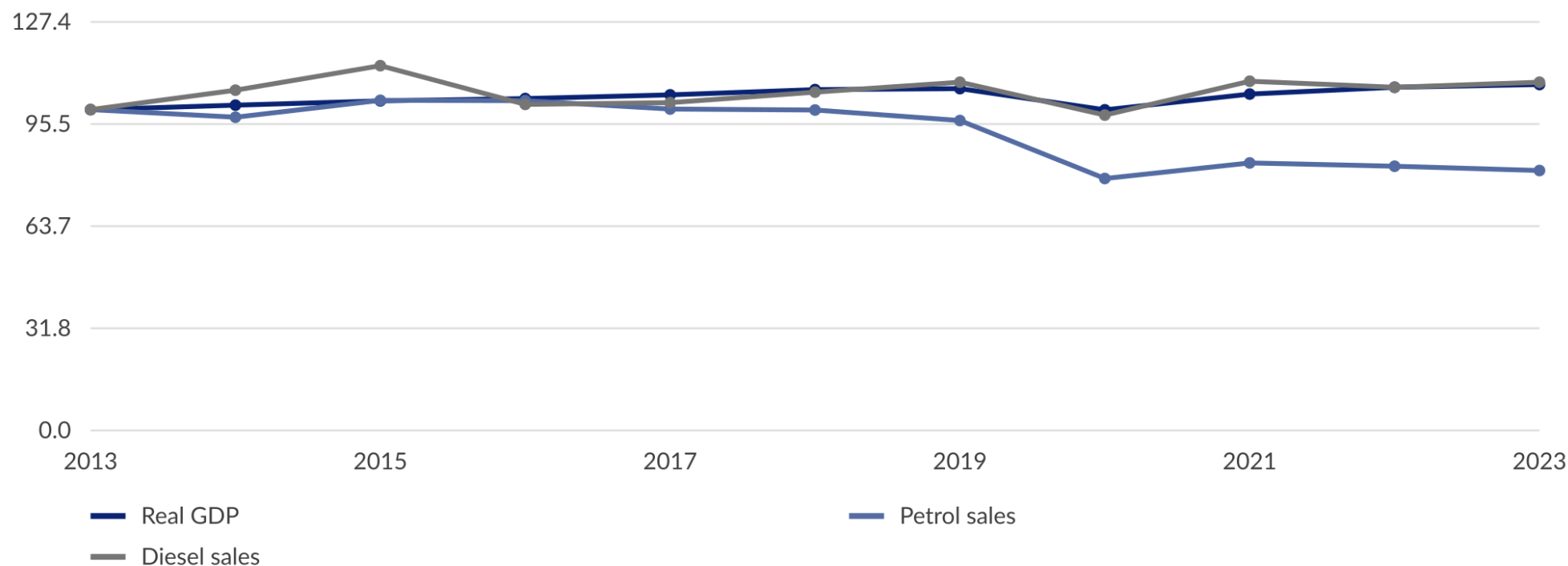
# The supporting evidence calibrates demand, supply and the terminal investment tests

| Evidence group         | Detail retained                                                   | Connection to the story                                          |
|------------------------|-------------------------------------------------------------------|------------------------------------------------------------------|
| Demand                 | Activity, freight, EVs, efficiency and power                      | D1–D5 create the three demand paths.                             |
| Supply                 | Plant output, gas availability and restart timing                 | S1–S4 create the three domestic-supply paths.                    |
| Capture and operations | Routes, customer share, working tankage and turnover              | T1–T5 translate each world into terminal flows and constraints.  |
| Returns                | Rates, costs, capex and funding; separate EV adjacency economics. | F1 translates each investment option into cash flow and returns. |

The main story runs from the opening thesis to the investment tests. The added lever register and power/supply calibration pages record what remains to be populated.

# Fuel demand has diverged from GDP, so sector activity and fuel intensity need separate treatment

index: 2013 = 100; matched 2013-2023 window

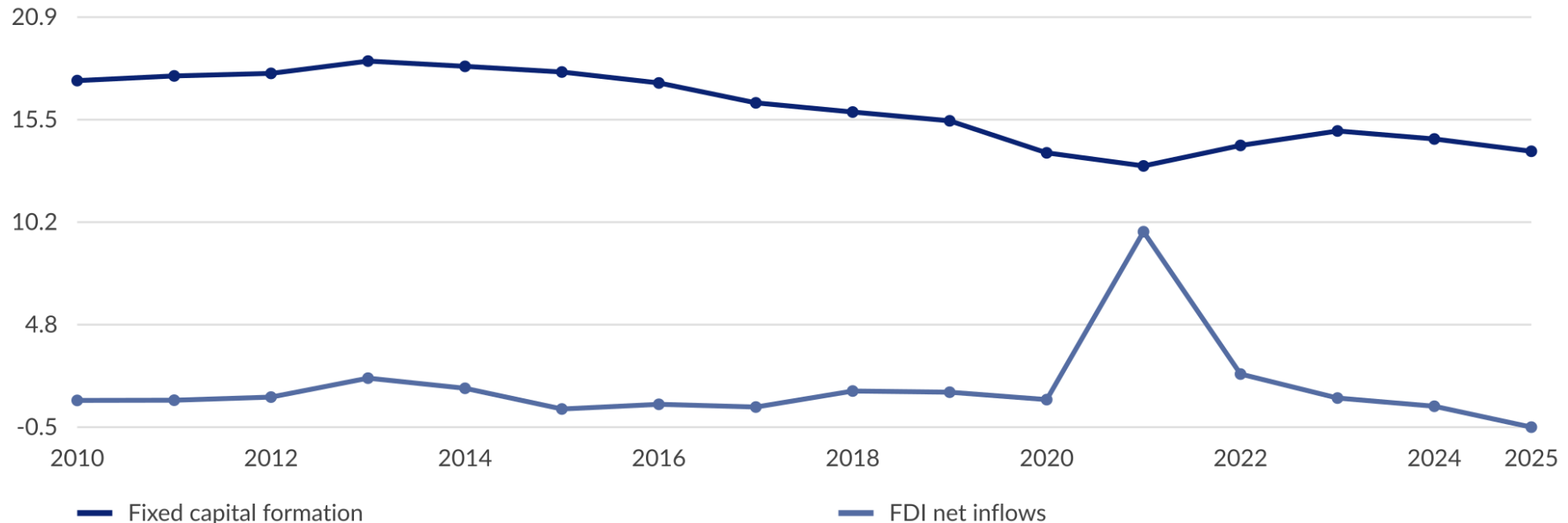


Economic weakness means limited growth in real activity and investment. Structural change means a different mix of output, travel and fuel use; this chart alone cannot attribute causality.

Staged GDP and department fuel sales, indices 2013=100. Sector attribution remains open; COVID distorts short-period comparisons.

# Fixed investment has weakened while FDI fluctuates, limiting a simple growth-led fuel argument

percent of GDP / calendar year



Fixed investment helps frame productive capacity. FDI can include large transactions; neither series is a direct investor-confidence measure or an additional fuel-demand multiplier.

World Bank WDI, FDI net inflows and gross fixed capital formation as % of GDP, 2010–2025. Confidence series remains an evidence gap.

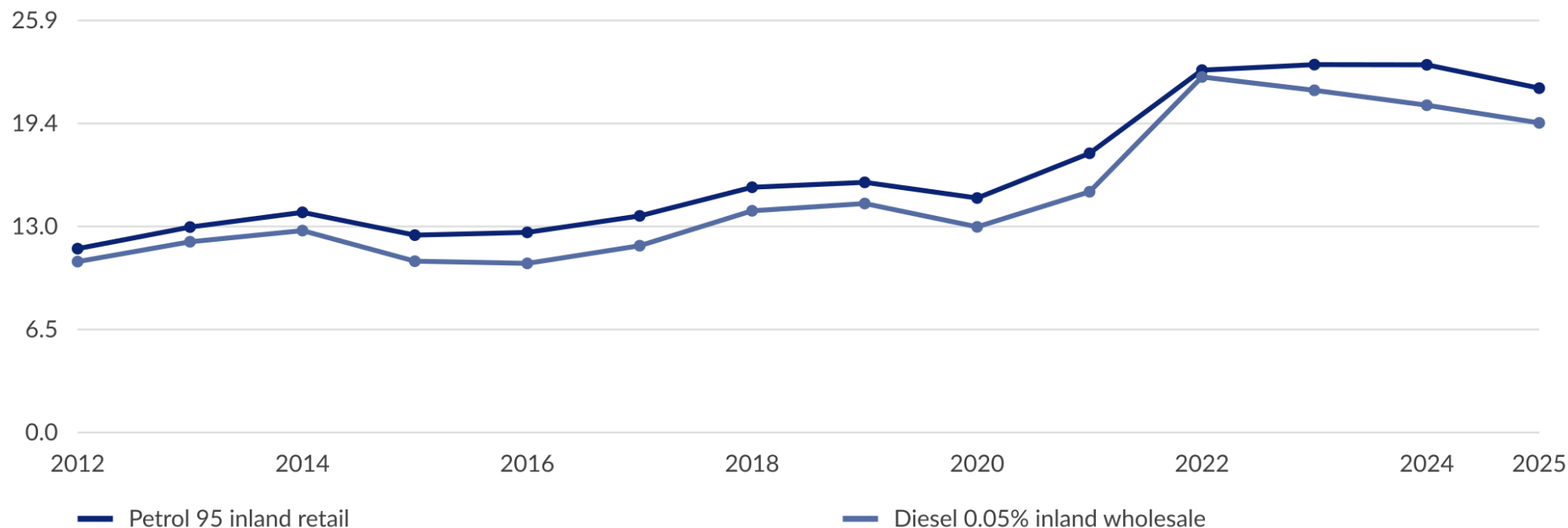
## Mining, agriculture and aviation expose fuel demand to different activity drivers

| Evidence             | Interpretation                                                                 | Implication for Vopak                                                                     |
|----------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Mining</b>        | Output, commodity mix and haul distance influence diesel use and freight.      | Separate mine activity from the fuel impact of changing freight mode.                     |
| <b>Agriculture</b>   | Crop cycles, weather, irrigation and logistics can shift seasonal diesel use.  | Establish sector activity and fuel exposure before assigning a headwind.                  |
| <b>Aviation</b>      | Traffic, distance, load factors and aircraft efficiency drive jet-fuel demand. | Keep jet fuel separate from petrol/diesel shares and verify terminal eligibility.         |
| <b>Manufacturing</b> | Real output and industrial composition help explain the economic backdrop.     | Test changes using sector data; a national fuel chart does not prove deindustrialisation. |

Analytical framework following user comments. Sector-specific exhibits and causal attribution remain to be completed.

# Higher fuel prices strengthen EV running-cost incentives, but ownership economics determine uptake

nominal ZAR/litre; full-year monthly averages



Shipping disruption, crude prices and FX can move fuel costs temporarily. EV adoption depends on sustained total ownership cost, purchase affordability, charging access and fleet replacement.

Department fuel-price history, nominal R/litre. No quantified geopolitical premium or calibrated EV adoption response is asserted.

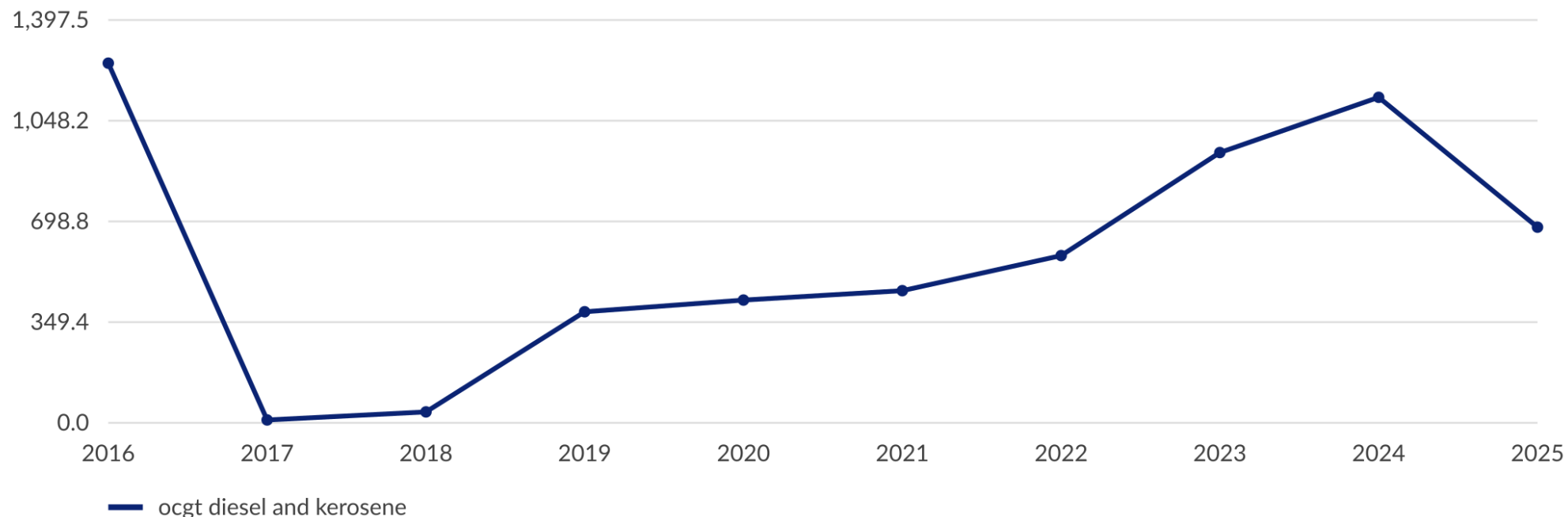
# New-vehicle efficiency gains affect fuel demand gradually as older vehicles leave the fleet

| Evidence        | Interpretation                                                                   | Implication for Vopak                                                                            |
|-----------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Fleet growth    | More vehicles or distance travelled can raise demand.                            | Measure activity before attributing lower fuel volumes to efficiency.                            |
| Efficiency      | New-cohort consumption gains enter the fleet over successive replacement cycles. | Obtain OEM and real-world evidence by segment; do not apply new-vehicle gains to every vehicle.  |
| EV substitution | EVs remove some combustion-engine vehicle activity from the fuel calculation.    | Count each vehicle transition once, then calculate efficiency on the remaining combustion fleet. |

Mechanism supported by existing cohort architecture. OEM historical and future efficiency evidence is still missing; no new forecast shown.

## Eskom's own OCGT fuel use fell 40% in FY2025 as availability improved

million litres / FY ending 31 March

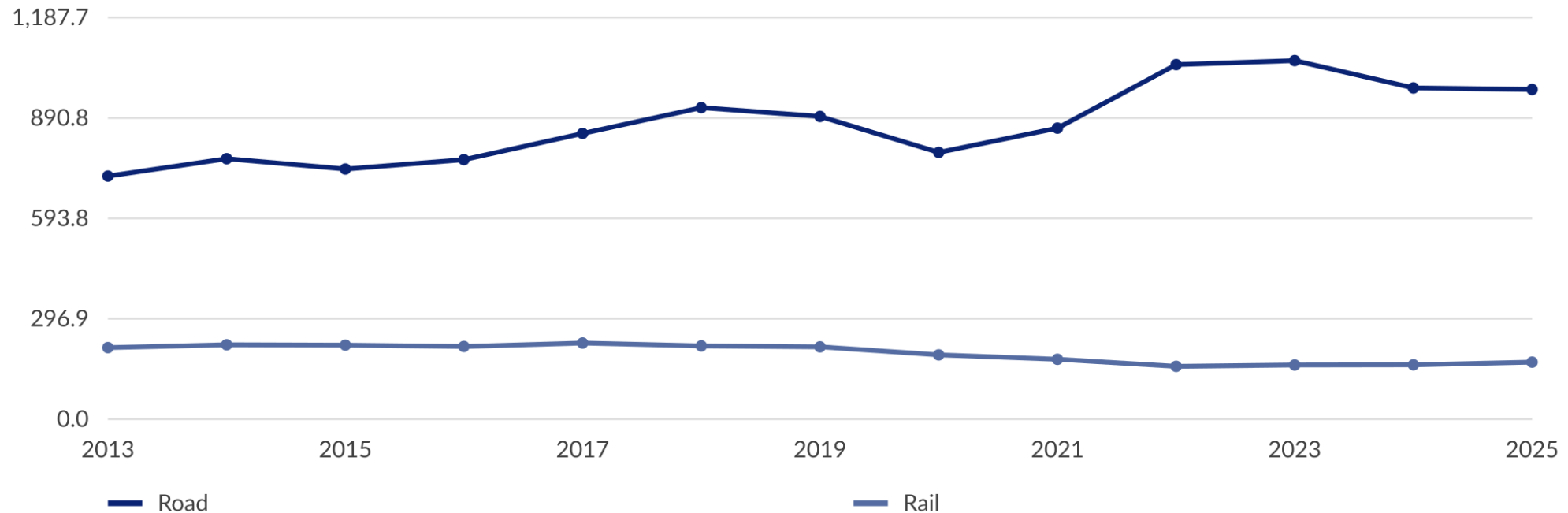


Reported fuel fell from 1,129.5 to 679.1 million litres as EAF improved from 54.56% to 60.60%. This removes part of the recent diesel-demand support.

Eskom integrated report 2025: FY March, own-fleet diesel + kerosene. EAF association is not a dispatch model; private backup use excluded.

# Road carries most reported freight tonnage, making transferable haul distance a key diesel lever

million tonnes / complete calendar year



Payload establishes the historical direction. Diesel displacement requires corridor tonne-kilometres, road fuel intensity, rail traction and last-mile activity.

Stats SA monthly freight payload, complete calendar years. National tonnes are not corridor tonne-kilometres or transferable freight.

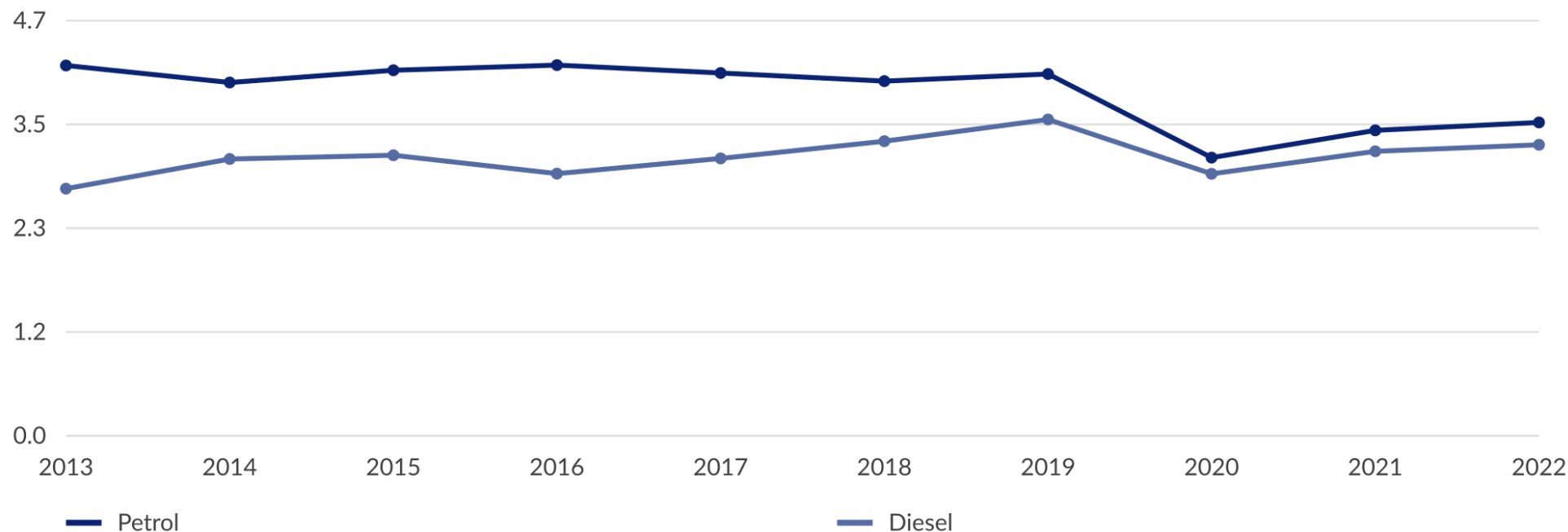
# Rail funding and reliable service determine how much road freight can return to rail

| Evidence                  | Interpretation                                                                                         | Implication for Vopak                                                                                       |
|---------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Network ambition</b>   | The supplied brief reports a 250 Mt target dependent on a funding plan.                                | Do not treat every additional rail tonne as road displacement. Commodity, corridor and origin matter.       |
| <b>Funding constraint</b> | Brief: R46.34bn approved over five years, including R1.71bn expansion.                                 | Maintenance and sustaining spend dominate. A published plan is insufficient evidence of delivered capacity. |
| <b>Container corridor</b> | Brief: R2.11bn in 2026/27–2028/29; private funding expected thereafter.                                | Prioritise Durban–Gauteng road/rail competition and reconcile the corridor to Network Statement v4.         |
| <b>Two fuel effects</b>   | Rail can displace truck diesel while improved infrastructure also changes fuel-distribution economics. | Keep freight-demand effects separate from the route delivering fuel to Gauteng.                             |

User-supplied AIA Transnet Funding Brief, Sep 2026, pp1–2; cites FY2026 AFS and Network Statement v4 Annexure 17. Primary figures not independently reconciled.

## Gauteng demand establishes market scale, while Lesedi capture and turnover determine throughput

billion litres / complete calendar year



2022 petrol + diesel sales total 6.81bn litres. Lesedi's 140,000m<sup>3</sup> equals 2.06% of that annual market per effective gross-capacity turn. Actual turns and Gauteng allocation remain unknown.

Department provincial sales, latest complete staged year 2022; Vopak Lesedi current capacity. Cross-vintage scale test only, not current observed market share.

# Every market world passes through explicit demand, supply, capture and operating levers

| Lever IDs               | Inputs to calibrate                                                                    | Output / remaining evidence                                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>D1–D5 demand</b>     | D1 sector growth; D2 rail; D3 EV adoption; D4 efficiency; D5 power.                    | Historical volumes and time series feed L/M/H demand. OEM, corridor tonne-km and backup evidence remain open.                     |
| <b>S1–S4 supply</b>     | S1 SAPREF; S2 Secunda gas/MRG; S3 Natref; S4 PetroSA.                                  | Dates, utilisation, availability and yields produce L/M/H domestic output. Medium and plant marginal effects require calibration. |
| <b>T1–T2 capture</b>    | T1 delivered route economics/access; T2 customer/product/destination capture.          | Allocate domestic and imported fuel to unique terminal deliveries. Tariffs, rights and commercial allocation remain open.         |
| <b>T3–T5 operations</b> | T3 eligible working tankage; T4 turnover; T5 receipt/dispatch, peaks and stock policy. | Compare required with achievable turns, occupancy and headroom. Existing 1/2/3-turn sensitivity is illustrative only.             |
| <b>F1 finance</b>       | Storage/handling terms, costs, capex, tax, working capital and hurdle rate.            | Translate constrained throughput into cash flow, NPV, IRR and investment triggers. Commercial inputs remain open.                 |

Medium is the reference market world; operational sensitivities are separate. Owner: Nigel / analyst and commercial teams. Resolve each missing input before numeric adoption or investment reliance.

# Generation timing must close the power balance before residual diesel can be forecast

| Time series               | Assumption / lever                                                                                      | Evidence status and next use                                                                                                  |
|---------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Recovery history          | EAF, unserved energy / load-shedding hours, OCGT GWh and litres.                                        | EAF, Eskom fuel and Eskom/IPP GWh staged. Comparable load-shedding and private backup series still needed.                    |
| Generation supply by year | Existing fleet and retirement profile; wind, solar and gas commissioning; availability and utilisation. | Obtain dated Eskom / approved planning schedules. Report annual GWh alongside dependable capacity and dispatch constraints.   |
| Residual dispatch         | Electricity demand, hourly/seasonal shape, storage, imports, reserve needs and fuel constraints.        | An annual GWh shortfall alone cannot determine diesel use. Calibrate feasible peaking dispatch, then apply consumption rates. |
| L/M/H demand contribution | Sustained recovery, reference and setback assumptions across the same planning horizon.                 | Output generation and diesel time series with solid actuals / dashed assumptions. No uncalibrated future values are drawn.    |

Requested generation-balance exhibit remains quantitatively incomplete. Owner: Nigel / power analyst; trigger: before D5 adoption. Do not convert installed renewable MW directly into dependable diesel displacement.

# SAPREF and Natref define capacity sensitivities; plant output must complete the supply paths

| Plant lever       | Required time-series inputs                                                                                    | Scenario treatment                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>S1 SAPREF</b>  | High illustration adds 400 kbpd: assumed 2029 FID plus 48 months gives 2033 availability. Low/Medium add none. | Separate the import phase from refinery commissioning; establish ramp-up, availability and product yields.                          |
| <b>S3 Natref</b>  | 108 kbpd retained in Medium/High; excluded in the Low stress case at 2036. No closure date assumed.            | This is a hypothetical availability test. Confirm operating outlook; do not infer closure from an outage.                           |
| <b>S2 Secunda</b> | 150 kbpd reported crude-equivalent footprint held in all three capacity illustrations.                         | Gas decline and MRG diversion can reduce liquid output without changing nameplate. Quantify the gas balance and liquids penalty.    |
| <b>S4 PetroSA</b> | No restart addition included in these three capacity illustrations.                                            | Test funded restart scope, feedstock, timing and yields separately; zero reported contribution is not proof of dismantled capacity. |

Illustrative footprint cases: 250 / 358 / 758 kbpd; not a complete low/base/high production range. Owner: Manish; Nigel review. Resolve dates, operating factors, yields and gas effects before nine-world output adoption or investment use.

# Public capacity ratios establish scale but cannot substitute for realised terminal throughput

| Catchment              | Initial inference                                                                                                         | What the result can support                                                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Durban                 | 45.1% of the combined published gross capacity of Vopak and Bidvest in Durban.                                            | A two-operator footprint proxy with mixed products and an incomplete market denominator. It does not establish Durban throughput share.         |
| Gauteng / Lesedi       | 2022 petrol + diesel sales: 6.81bn litres. Lesedi's gross capacity equals 2.06% of that market per effective annual turn. | A cross-vintage scale test. Current sales, eligible working capacity, actual turns and Gauteng allocation are needed for a throughput estimate. |
| Investment implication | Convert eligible tankage and realised turns into unique customer deliveries, then compare with existing capacity.         | A large market can coexist with spare terminal capacity. Market scale alone does not establish a need for additional tanks.                     |

Operator capacities: Vopak Durban 360,246m³, Bidvest Durban 439,306m³, Lesedi 140,000m³. Department Gauteng sales 2022; appendix p27. Transfers between terminals must not be double-counted.

## A matched demand snapshot is available, but the domestic supply balance still needs reconciliation

| Evidence / bn litres    | Petrol                                                                                                    | Diesel                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 2022 national sales     | 9.18                                                                                                      | 12.72                                                                                      |
| 2022 Gauteng sales      | 3.53                                                                                                      | 3.28                                                                                       |
| 2022 imports / exports  | 5.46 / 0.85                                                                                               | 11.95 / 0.89                                                                               |
| 2021 balance diagnostic | 0.51 statistical difference needed to reconcile reported supply and final consumption.                    | 2.10 statistical difference needed to reconcile reported supply and final consumption.     |
| Decision consequence    | Sales less net imports is a residual, not measured domestic output. Confirm product and stock boundaries. | Provincial sales establish demand scale, not source allocation or flows captured by Vopak. |

Department sales: four quarters; SARS trade: 12 months and litre units only, blends/other units excluded. 2021 energy-balance consumption differs from sales; statistical differences are not stock changes. Source hashes and exact values: [investment\\_bridge\\_2026\\_10\\_08/evidence.json](#).

## The six-step bridge is implemented for screening; calibration now depends on the requested inputs

| Work package                  | Implemented now                                                           | Input needed to complete the decision                  |
|-------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------|
| <b>1   Baseline</b>           | Matched 2022 sales/trade; 2021 balance diagnostic.                        | Reconcile production, stocks and source definitions.   |
| <b>2   Route competition</b>  | Delivered cost and bearable-charge calculation; route readiness register. | Matched quotes, service capacity and access.           |
| <b>3   Vopak capture</b>      | Capture calculation constrained by accessible market and route capacity.  | Customer allocation and unique terminal deliveries.    |
| <b>4   Existing assets</b>    | Required-turn and operating-bottleneck calculation; gross-turn screen.    | Working capacity, achievable turns and peak limits.    |
| <b>5   Investment options</b> | Pre-tax break-even volume calculation.                                    | Commercial inputs and full incremental cash flows.     |
| <b>6   Nine worlds</b>        | Nine-cell output register and demand/supply balance calculation.          | Annual L/M/H paths; rerun routes, capture and returns. |

Screening functions tested with synthetic cases; commercial outputs remain unpopulated. Not integrated into the forecast engine or independently reviewed. Owners, request IDs and expiry triggers: [workstreams/WSO\\_governance/workplan/sa\\_investment\\_bridge\\_2026\\_10\\_08.csv](workstreams/WSO_governance/workplan/sa_investment_bridge_2026_10_08.csv).

# Demand, supply and corridor access translate into captured throughput and required tank turns

| Argument                                                                                                      | Situation                                                                                     | Complication                                                                                   | Resolution / decision                                                                                        |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  <b>Transport demand</b>     | Subdued aggregate growth masks different product trends. Road freight supports diesel use.    | Sector growth, rail recovery, EVs and efficiency change fuel demand.                           | D1–D4 define activity, rail, EV and efficiency paths. Translate them into capturable product demand.         |
|  <b>Power demand</b>         | Power shortages supported diesel use; Eskom consumption has recently fallen.                  | Recovery may persist or reverse through outages and replacement delays.                        | D5 adds fleet recovery, replacement timing and diesel dispatch to the same demand paths.                     |
|  <b>Domestic supply</b>      | Reduced local production has increased reliance on imports.                                   | SAPREF, Natref and Secunda can change domestic fuel output materially.                         | S1–S4 define plant output and sourcing mix. Imports are one consequence of the domestic balance.             |
|  <b>Corridor competition</b> | Durban connects imported fuel to inland customers.                                            | Alternative gateways, delivered costs, reliability and access alter flows.                     | T1–T2 compare delivered economics and capture for domestic and imported fuel serving common destinations.    |
|  <b>Vopak investment</b>   | Durban and Lesedi provide connected assets whose throughput can rise through higher turnover. | Working tankage, receipt/dispatch, peaks and stock requirements constrain achievable turnover. | T3–T5 compare working capacity, required/achievable turns and dispatch limits. F1 tests incremental returns. |

Lever IDs are mapped in the appendix. All nine market worlds use explicit routing, capture and operational assumptions. Tank occupancy, turnover and throughput utilisation are separate measures. Competitiveness varies by destination; neither alternative gateway is yet established as competitive into Gauteng.

## The illustration gives extra tanks only R8m more NPV than improvement alone

| High demand / low supply  | Capital and throughput                           | Incremental value versus existing                                     |
|---------------------------|--------------------------------------------------|-----------------------------------------------------------------------|
| Existing                  | R0m capex<br>1.890bn L/year served               | NPV R+0.0m<br>Unserved 0.923bn L/year                                 |
| Improve access / dispatch | R100m capex<br>2.268bn L/year served             | NPV R+27.0m<br>Unserved 0.545bn L/year                                |
| Improve + add tanks       | R600m capex<br>2.813bn L/year served             | NPV R+35.1m<br>Unserved 0.000bn L/year                                |
| Decision threshold        | Improvement beats existing above 2.211bn L/year. | Tanks beat improvement above 2.806bn L/year; assumed capture 2.813bn. |

ILLUSTRATIVE ONLY. Combined package = R100m access/dispatch improvement + R500m for a further 40,000 gross m<sup>3</sup> beyond the reported 140,000 m<sup>3</sup>. Alternatives are authored packages, not engineered designs. Full incremental pre-tax cash flows in delivered CSV; 100% site basis. NPV uses 10% and 20 operating years; commercial and delivery risks uncalibrated.

# Vopak reports anchor capacity and the revenue model; they do not disclose Lesedi project economics

| Evidence class                | What the sources establish                                                                                      | How it is used                                                                                                    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>REPORTED   Assets</b>      | Lesedi: 140,000 m <sup>3</sup> ; Durban: 360,246 m <sup>3</sup> . Vopak share: 70% at each site.                | Use full physical capacity for operations. Annual report confirms Lesedi expansion was commissioned in 2025.      |
| <b>REPORTED   Revenue</b>     | Storage services: EUR1,058.8m of EUR1,298.9m consolidated revenue in 2025.                                      | 81.5% is storage revenue. Rental may include minimum throughput; extra turns are not automatically extra revenue. |
| <b>INFERRED   Group ratio</b> | EUR93.9 revenue / gross m <sup>3</sup> /year: proportional revenue divided by year-end proportional capacity.   | Mixed-portfolio scale indicator only. Not a Lesedi tariff, margin or estimate of site revenue.                    |
| <b>REPORTED   Benchmarks</b>  | 15.6% group operating cash return; 5–7x capex/EBITDA for gas/industrial and 4–8x for transition infrastructure. | Context, not South Africa fuel-project hurdle rates. Authored 10% discount rate is separate.                      |

Vopak Annual Report 2025 pp36/39, 63, 213 and 324; FY2025 Analyst Presentation p24; official terminal directory checked 8 Oct 2026. All other business units includes multiple countries; do not allocate its profit to South Africa. URLs, downloaded originals and hashes retained.

## The investment result rests on explicit assumptions that must be replaced or agreed

| ILLUSTRATIVE input       | Value used                                                                                                   | What must be validated                                                                                                     |
|--------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Catchment balance</b> | Demand 6.0 / 6.8 / 7.6bn L; domestic deliveries 2.4 / 3.6 / 4.8bn L.                                         | Steady-state stress points; M rounds 2022 Gauteng sales for scale. No national production forecast implied.                |
| <b>Capture</b>           | 70% of imports via Durban; Vopak captures 80%; 90% goes to Lesedi; 8% domestic capture.                      | No observed shares. Capture is unique Lesedi flow; Durban transfers are not added to market demand.                        |
| <b>Operations</b>        | 90% working fraction; turns 1.25 existing / 1.50 improved; +40,000 gross m <sup>3</sup> tanks.               | 126,000 working m <sup>3</sup> initially; 162,000 with tanks. Assumed receipt/dispatch ceiling 3.5bn L/year.               |
| <b>Revenue</b>           | R1,500 / gross m <sup>3</sup> /year rent on added contracted tanks; R60 / extra m <sup>3</sup> contribution. | Chargeable excess services only, excluding rent/minimum-throughput inclusions. Added rent follows incremental storage use. |
| <b>Capital and costs</b> | R100m improvement + R500m tanks; fixed opex R5m + R12m/year; maintenance 1% capex/year.                      | Author-selected cost sensitivities; no Vopak quotation or engineering estimate.                                            |
| <b>Cash flow</b>         | 20 years at 10%; 50% / 75% / 100% operating ramp; capex upfront.                                             | Pre-tax unlevered constant ZAR. Tax, working capital, construction timing and residual value excluded.                     |

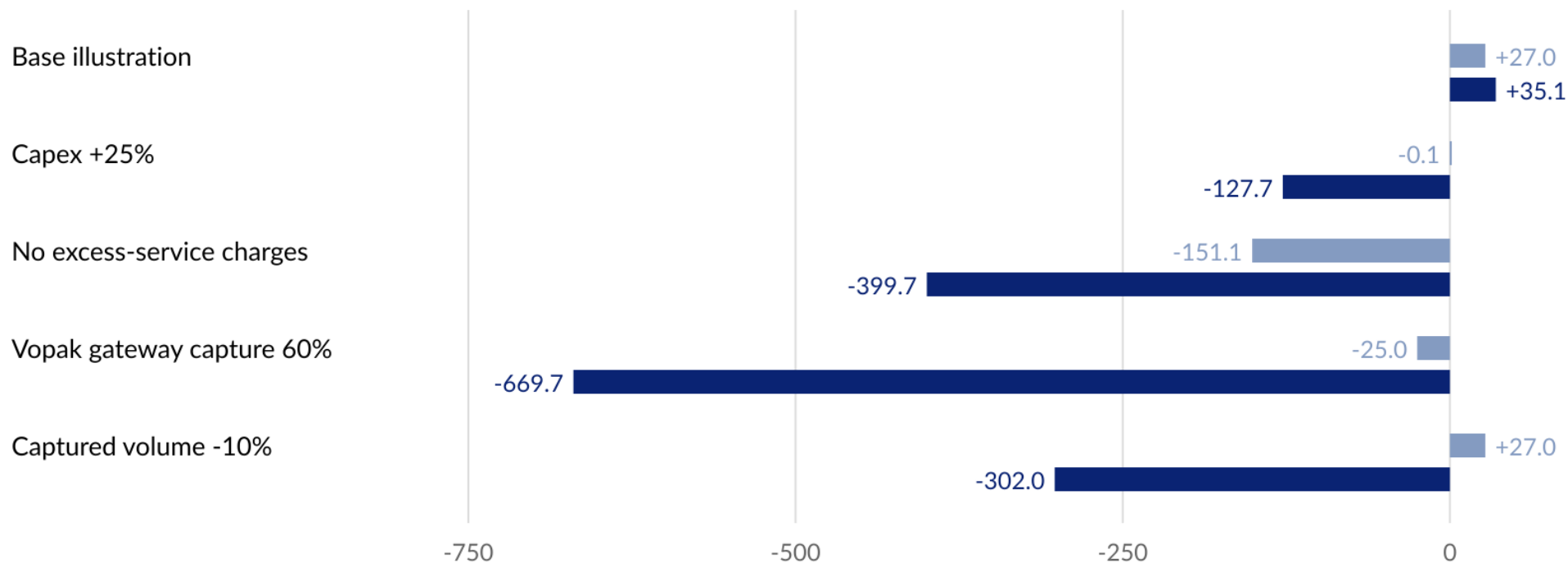
Every illustrative value is registered in assumptions/2026/vopak\_investment\_illustration.yaml. Not a forecast, commercial offer or approval. Proposed validation: Nigel coordinates Vopak commercial/operations; Manish reconciles; independent review outstanding.

# Small changes in capture or economics overturn the illustrated tank-expansion choice

## ILLUSTRATIVE incremental NPV, Rm

Improvement

Improvement + tanks



The illustration supports validating contracts and a low-capex improvement first. Extra tanks have only a narrow advantage and fail the tested downside cases.

ILLUSTRATIVE high-demand/low-domestic-delivery case. NPVs are incremental versus existing assets at 10%, pre-tax. Each shock applied separately: capex +25%; no chargeable excess services; Vopak gateway capture 80%→60%; captured Lesedi flow -10%. No probabilities. Full numeric outputs and cash flows retained.

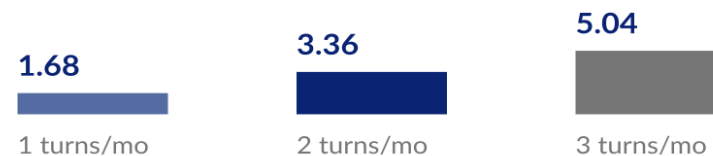
# Illustrative tank turns test handling potential; actual operating limits still need validation

## Gross handling sensitivity

### Durban



### Lesedi



bn litres/year; 1 / 2 / 3 monthly turns are an existing authored sensitivity, not measured site limits.

## Investment test

### 01 | Calculate required turns

Captured annual throughput divided by eligible working capacity, then by 12.

### 02 | Test achievable handling

Check receipt / dispatch, pipeline access, seasonal peaks and stock policy.

### 03 | Identify the binding constraint

Use existing assets first; debottleneck or add tanks where the constraint is profitable.

### Next action

Confirm working tankage and achieved turns before sizing any additional storage.

Required working-capacity turns must be compared with achievable turns at each terminal. Gross handling equivalents do not establish spare capacity or actual market share.

Registered REG-TERMINAL-HANDLING-TURNS: 1/2/3 turns per month, authored operational sensitivity, not the demand-supply worlds or measured limits. Gross Durban mixed-product tankage; no assumed eligible fraction. Do not sum inter-terminal transfers as unique demand. Separate p4/p5 illustration assumes 90% working capacity and 1.25/1.50 turns; neither is inferred from group occupancy.